

KRASNY YAR, Russian Federation

WMO: 318450

Lat: 46.5404N

Lon: 135.3357E

Elev: 426

StdP: 14.47

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.0	-30.1	-39.0	0.6	-33.8	-35.0	0.8	-29.7	13.4	9.0	11.1	6.0	0.1	80	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	87.5	72.1	84.6	71.3	81.7	69.8	75.6	83.2	73.6	81.4	71.9	78.9	3.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	125.7	79.9	71.3	117.3	77.6	69.6	110.4	76.0	39.5	83.3	37.6	81.5	36.1	79.1	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	9.7	8.2	DB	-39.5	93.0	4.8	2.2	-42.9	94.6	-45.7	95.9	-48.4	97.2	-51.9	98.8	(4)
(5)			WB	-39.7	79.3	4.8	2.4	-43.1	81.0	-45.9	82.4	-48.6	83.7	-52.0	85.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	-6.7	1.6	18.5	38.5	53.5	63.7	70.6	68.2	55.9	39.5	17.2	-2.6	(6)
(7)		DBStd	28.46	9.30	9.54	11.35	7.96	6.78	6.13	4.50	5.31	6.84	7.99	12.22	10.07	(7)
(8)		HDD50	7453	1756	1354	975	353	43	0	0	0	23	334	985	1630	(8)
(9)		HDD65	11322	2221	1774	1440	794	360	95	9	30	280	789	1435	2095	(9)
(10)		CDD50	1982	0	0	0	9	151	411	637	563	201	10	0	0	(10)
(11)		CDD65	377	0	0	0	0	3	56	182	128	8	0	0	0	(11)
(12)		CDH74	3515	0	0	0	23	211	811	1480	890	97	3	0	0	(12)
(13)		CDH80	995	0	0	0	4	44	254	477	204	12	0	0	0	(13)
(14)	Wind	WSAvg	2.7	2.0	2.4	3.3	3.4	3.4	2.5	2.2	2.1	2.3	2.9	3.1	2.3	(14)
(15)	Precipitation	PrecAvg	34.3	0.7	0.7	1.4	1.9	3.2	3.7	6.9	5.9	4.1	2.8	1.8	1.3	(15)
(16)		PrecMax	53.6	1.5	1.3	3.2	3.4	6.3	9.4	15.7	11.1	6.8	6.9	2.9	3.0	(16)
(17)		PrecMin	24.1	0.1	0.1	0.1	0.6	0.8	1.7	1.7	1.5	1.5	0.7	0.5	0.3	(17)
(18)		PrecStd	5.7	0.4	0.4	0.7	0.8	1.4	1.9	3.2	2.4	1.5	1.3	0.6	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.1	37.3	52.9	77.1	85.2	90.5	91.6	88.8	81.4	71.2	52.1	30.4	(19)
(20)			MCWB	23.0	31.5	40.9	55.1	64.1	71.6	74.7	74.9	69.0	55.7	42.6	28.1	(20)
(21)		2%	DB	19.0	31.1	45.9	68.7	79.6	86.2	88.2	84.8	76.5	64.3	44.1	25.7	(21)
(22)			MCWB	16.9	26.4	36.6	50.4	60.3	69.6	73.5	73.7	65.4	52.6	38.5	23.9	(22)
(23)		5%	DB	14.2	25.6	41.4	61.1	75.3	82.7	85.3	82.0	72.9	59.2	39.3	20.3	(23)
(24)			MCWB	12.4	22.1	33.8	46.4	58.1	67.9	72.6	71.9	63.1	49.1	35.5	18.6	(24)
(25)		10%	DB	10.4	21.0	37.5	55.7	70.6	78.5	82.1	79.1	69.3	54.5	34.6	14.9	(25)
(26)			MCWB	8.9	17.9	31.9	43.8	55.9	66.3	71.5	70.7	61.4	46.9	31.9	13.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.3	32.2	41.6	56.0	64.8	74.4	78.1	78.0	71.0	57.4	44.1	28.7	(27)
(28)			MCDB	25.7	36.4	50.8	75.7	81.4	86.5	85.8	84.8	77.4	66.9	50.5	30.0	(28)
(29)		2%	WB	16.9	26.5	37.5	51.3	62.2	71.2	76.0	75.5	67.5	54.1	39.3	24.0	(29)
(30)			MCDB	18.7	30.9	44.0	65.7	75.6	83.0	84.1	81.9	73.8	61.3	43.3	25.6	(30)
(31)		5%	WB	12.6	22.1	34.8	47.7	59.8	69.2	74.1	73.6	64.9	51.0	35.8	18.6	(31)
(32)			MCDB	14.1	25.6	40.5	59.2	71.8	79.9	82.4	80.0	70.7	57.4	39.0	20.2	(32)
(33)		10%	WB	8.8	17.9	32.0	44.6	57.4	67.2	72.5	71.8	62.4	47.7	32.1	13.5	(33)
(34)			MCDB	10.3	21.0	37.4	53.8	67.6	76.7	80.2	77.2	67.7	53.3	34.6	14.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	29.2	33.5	29.5	26.0	26.8	24.3	20.2	19.1	24.1	25.1	22.7	25.8	(35)
(36)			MCDBR	30.8	36.7	32.0	39.4	38.4	33.4	27.4	24.1	30.0	34.8	27.0	24.7	(36)
(37)			MCWBR	28.6	31.6	23.0	22.6	19.5	17.3	14.2	13.7	18.5	22.1	20.2	23.3	(37)
(38)		5% WB	MCDBR	30.5	35.3	28.4	35.8	32.7	29.1	23.1	20.3	24.7	30.2	24.4	24.0	(38)
(39)			MCWBR	28.3	30.6	20.9	21.3	17.7	16.0	13.3	12.6	16.1	20.3	18.9	22.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.312	0.369	0.451	0.549	0.578	0.545	0.512	0.472	0.428	0.440	0.393	0.334	(40)	
(41)		taud	2.039	1.877	1.731	1.705	1.732	1.903	2.048	2.162	2.191	1.994	2.002	2.000	(41)	
(42)		Ebn at Noon	234	239	236	226	226	235	241	246	246	214	198	205	(42)	
(43)		Edh at Noon	31	46	62	70	71	60	51	44	39	41	33	30	(43)	
(44)	All-Sky Solar Radiation	RadAvg	633	989	1334	1416	1627	1747	1620	1442	1232	838	558	481	(44)	
(45)		RadStd	38	48	109	145	151	212	145	89	108	70	51	31	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page